



بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

تجرى بعونه تعالى مناقشة رسالة طالب الماجستير

محمود شمران عذيب

الموسومة بـ:



An Efficient Deep Learning Approach with Modified CNN for Multi-Class Brain Tumor Classification



القاعة المركزية للتعليم المستمر
جامعة واسط



الساعة التاسعة صباحاً
(09:00 AM)



يوم الأحد الموافق
2026/08/16

وتتألف لجنة المناقشة من الأساتذة المدرجة أسماؤهم أدناه

ت	اللقب العلمي	الاسم التدريسي	الشهادة	التخصص الدقيق	مكان العمل	المنصب
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2	أستاذ مساعد	أ.م.د. جعفر صادق قطيف	دكتوراه	شبكات الحاسوب	جامعة واسط / كلية التربية للعلوم الصرفة	عضواً
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تتشرف بحضوركم ومشاركاتكم في إنجاح هذه المناقشة العلمية

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

{ يَرْفَعِ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ
أُوتُوا الْعِلْمَ دَرَجَاتٍ }

صَدَقَ اللَّهُ الْعَظِيمُ

سورة المجادلة

الآية (11)

Republic of Iraq
Ministry of Higher Education and Scientific Research
Wasit University
College of Education for Pure Sciences
Department of Computer



An Efficient Deep Learning Approach with Modified CNN for Multi-Class Brain Tumor Classification

By: Mahmoud Shamran Atheeb

B.Sc. Computer — Wasit University (2022)

Supervised by: Dr. Baraa Ismeal Farhan | Assistant Professor

Presentation Outline

20-minute thesis defense — structure overview

01

Motivation & Problem Statement

Why brain tumor classification matters

02

Literature Review & Research Gap

State-of-the-art and identified gaps

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Proposed Architecture

Modified CNN with MAFE · MMSP · CAIP

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Methodology

Dataset, preprocessing & training protocol

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Results & Evaluation

Performance on Mendeley + 3 external datasets

06

Comparative Analysis

Against 9 competing algorithms

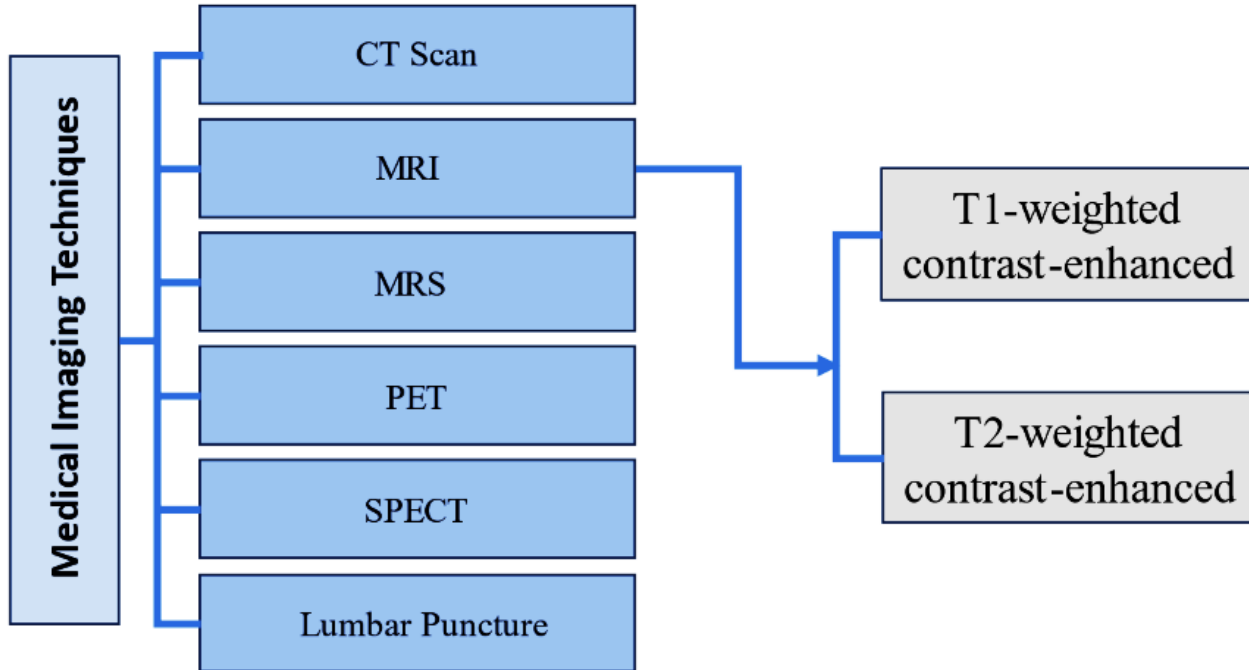
07

Conclusions & Future Work

Key contributions and next steps

Motivation & Problem Statement

The clinical and technical challenges driving this research



Motivation & Problem Statement

The clinical and technical challenges driving this research

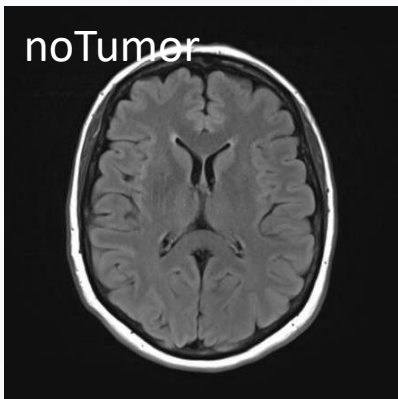
Glioma



Meningioma



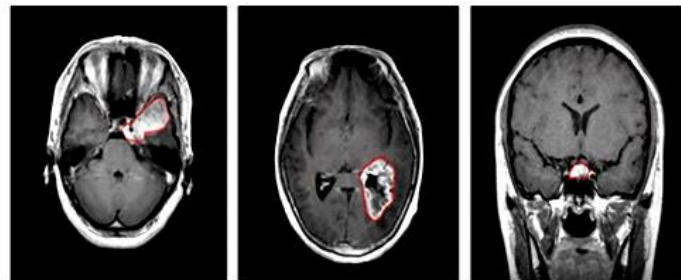
noTumor



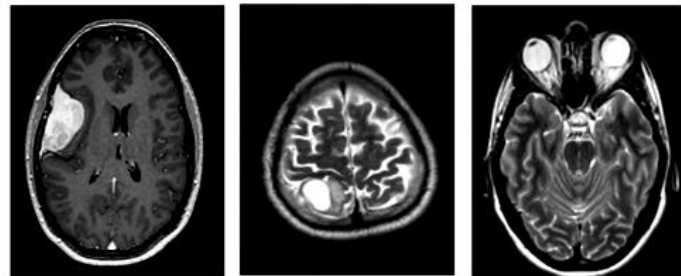
Pituitary



T1-weighted



T2-weighted



Meningioma

Glioma

Pituitary

Motivation & Problem Statement

The clinical and technical challenges driving this research

Clinical Challenge

- 300,000+ new brain & CNS tumor cases annually worldwide
- Brain tumors are the 2nd leading cancer-related death in adults under 40
- Intra-observer error among radiologists: up to 5–15% on complex cases
- Severe shortage of neuroradiologists in low-income countries

Technical Gaps

Computational Cost

VGG16: 138M params · ResNet50: 25.5M · Not mobile-deployable

Generic Extraction

Standard convolutions ignore clinically relevant MRI regions

Single-Scale Processing

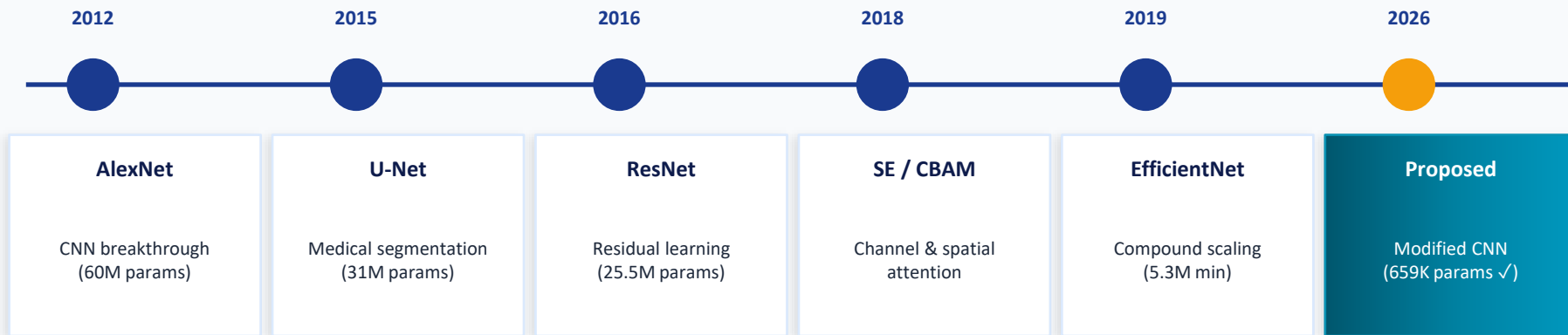
Fails to capture multi-scale tumor morphology

Static Pooling

Uniform pooling discards critical boundary features

Literature Review & Research Gap

Evolution of deep learning architectures for brain tumor classification

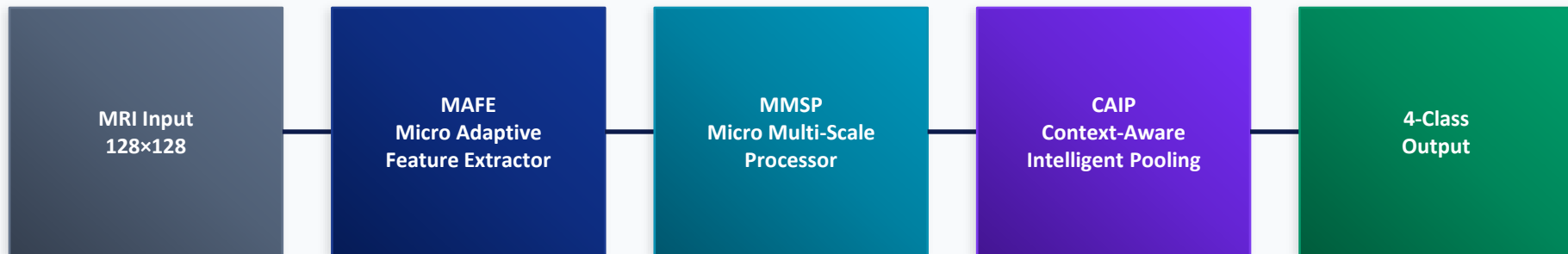


Identified Research Gap:

No existing lightweight architecture simultaneously achieves state-of-the-art accuracy AND computational efficiency for mobile clinical deployment. Existing models either sacrifice accuracy (SVM, XGBoost) or require impractical resources (VGG16, ResNet). The proposed Modified CNN bridges this gap through principled co-design of three orthogonal modules.

Proposed Architecture: Modified CNN

Three intelligent modules co-designed for brain tumor MRI classification



MAFE — Micro Adaptive Feature Extractor

- › Sequential spatial-then-channel attention (reverses CBAM ordering)
- › A tested parallel variant underperformed (99.17%), confirming sequential design
- › Enriches SE networks with joint spatial attention
- › Dynamically refines spatial and channel features

MMSP — Micro Multi-Scale Processor

- › Three-pathway multi-resolution feature fusion
- › Depthwise separable convolutions, single forward-pass block (vs. FPN's full encoder-decoder)
- › Simulates multi-resolution reasoning of expert radiologists

CAIP — Context-Aware Intelligent Pooling

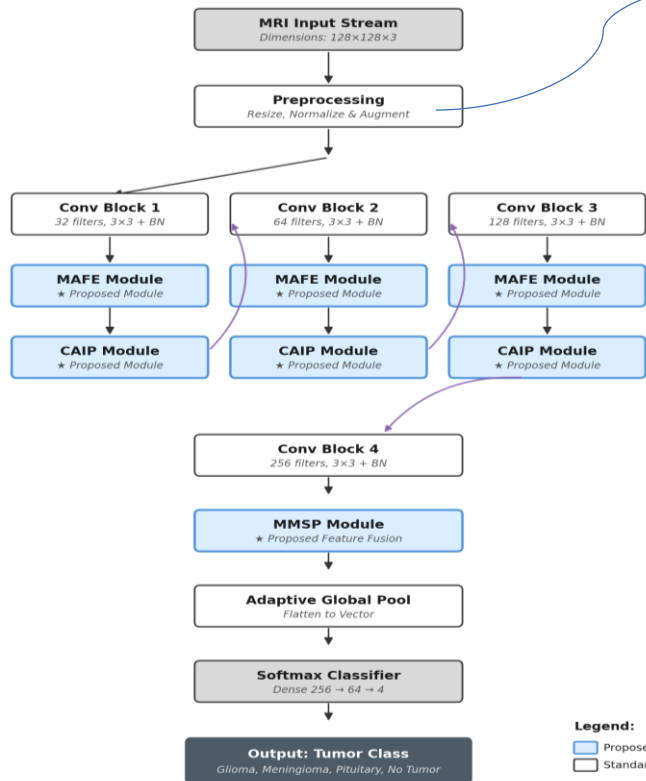
- › Sigmoid-gated content-adaptive pooling
- › Learns local optimal blend coefficients (not static)
- › Preserves diagnostically critical boundary features
- › Individually contributed +1.45% accuracy improvement (additive ablation)

Proposed Architecture: Modified CNN

Three intelligent modules co-designed for brain tumor MRI classification

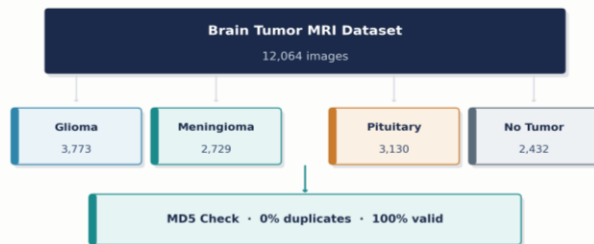
Ultra-Lightweight CNN Architecture

Brain Tumor MRI Classification – MAFE + MMSP + CAIP

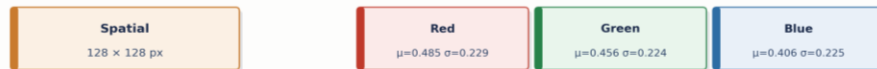


Dataset Preparation Overview

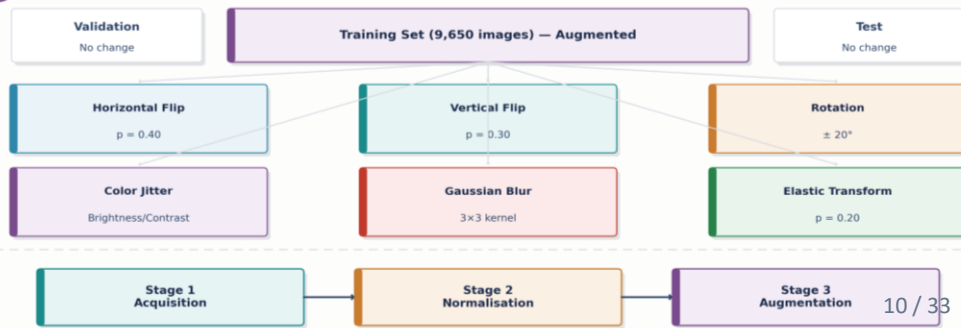
1 Data Acquisition



2 Normalisation



3 Augmentation



Dataset & Methodology

Mendeley Brain Tumor MRI — 4-class classification with 12,064 images

Primary Dataset: Mendeley Brain Tumor MRI

	Glioma ~25% of dataset	~31.28%
	Meningioma ~25% of dataset	~22.62%
	Pituitary Adenoma ~25% of dataset	~25.95%
	No Tumor (Normal) ~23% of dataset	~20.16%

Total: 12,064 high-resolution MRI images

4-Phase Methodology

Phase 1 **Dataset Preparation**
Spatial/intensity normalization, augmentation (flip, rotate, zoom)

Phase 2 **Architecture Design**
MAFE + MMSP + CAIP module co-design

Phase 3 **Training & Optimization**
Adam optimizer, LR scheduler, batch norm, dropout

Phase 4 **Evaluation & Validation**
Accuracy, F1, AUC, Grad-CAM, 3 independent runs

External Validation Datasets:

Kaggle Brain MRI · Br35H · Figshare

Testing generalization across independent repositories

Key Classification Results

Primary evaluation on Mendeley Brain Tumor MRI dataset

OVERALL ACCURACY

99.22

%

TRAINABLE PARAMS

659K

parameters only

INFERENCE LATENCY

12 ms

per image

MODEL SIZE

7.88 MB

on disk

RAM USAGE (Batch)

1.8 MB

memory footprint

PARAM EFFICIENCY

15.5%

more efficient vs U-Net

CAIP CONTRIBUTION

+1.45%

accuracy gain (ablation)

VS U-NET PARAMS

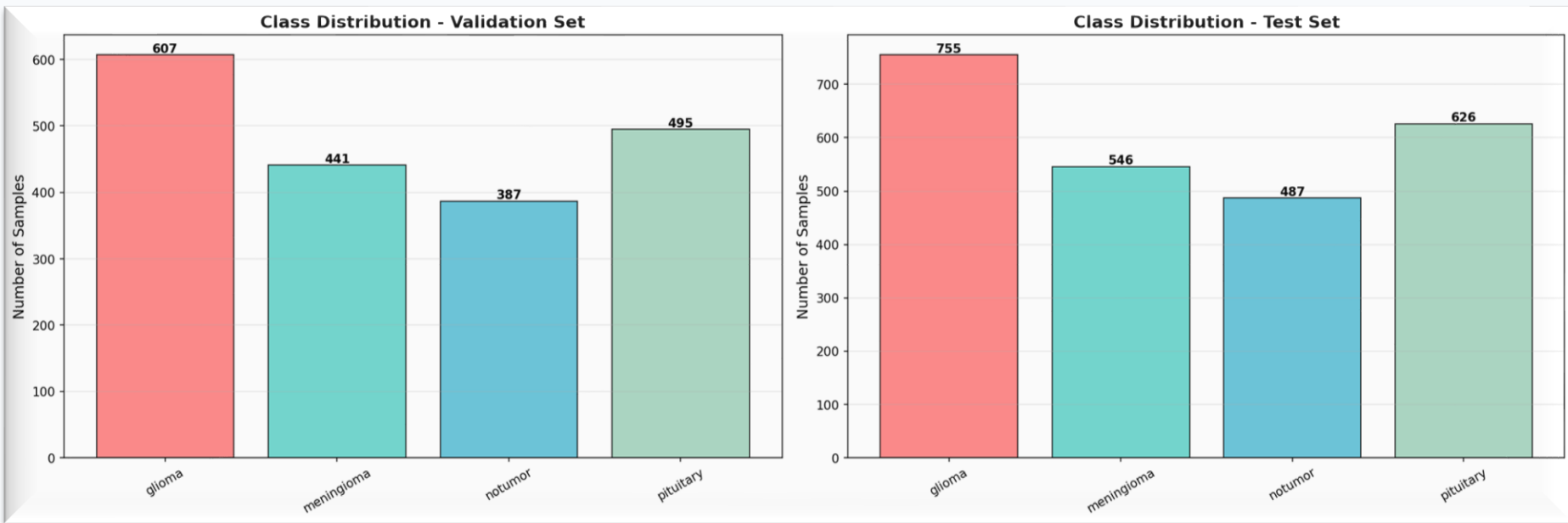
98%

fewer parameters

Tumor Class	Precision	Recall	F1-Score
Glioma	99.1%	98.8%	98.9%
Meningioma	99.2%	99.5%	99.3%
Pituitary	99.6%	99.4%	99.5%
No Tumor	99.4%	99.3%	99.4%

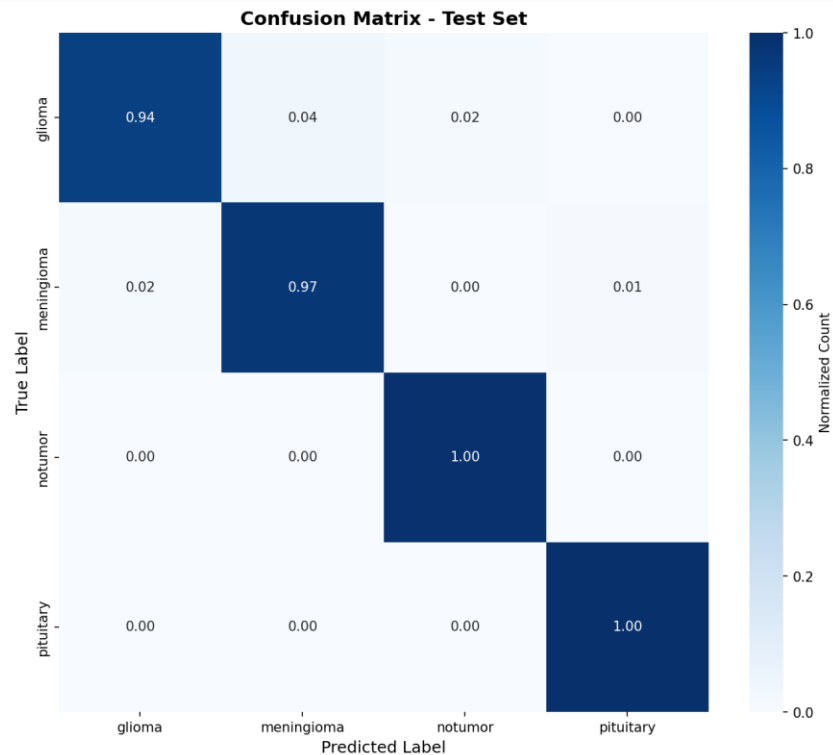
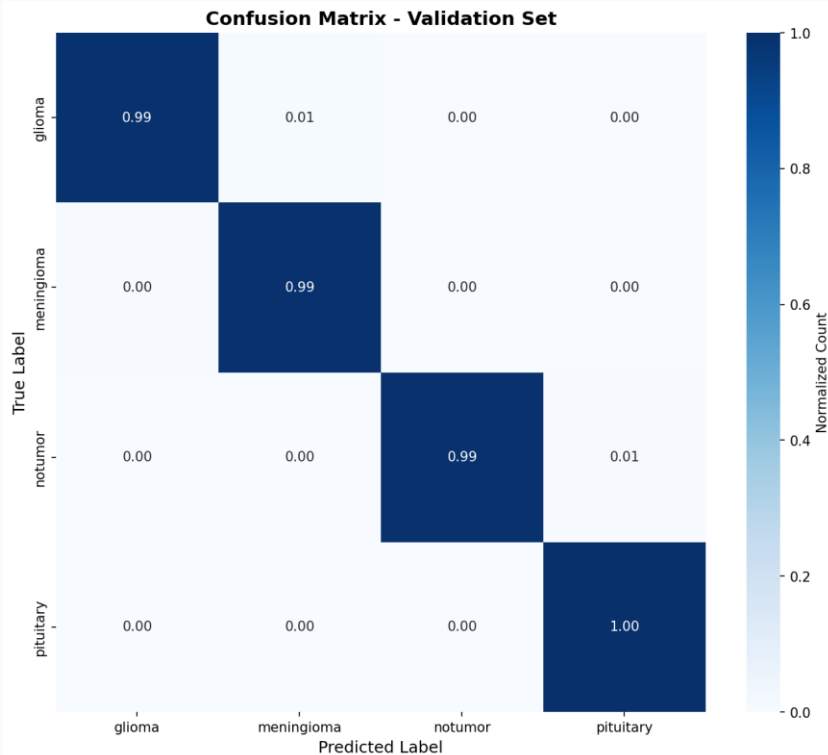
Test Vs Validation

Primary evaluation on Mendeley Brain Tumor MRI dataset



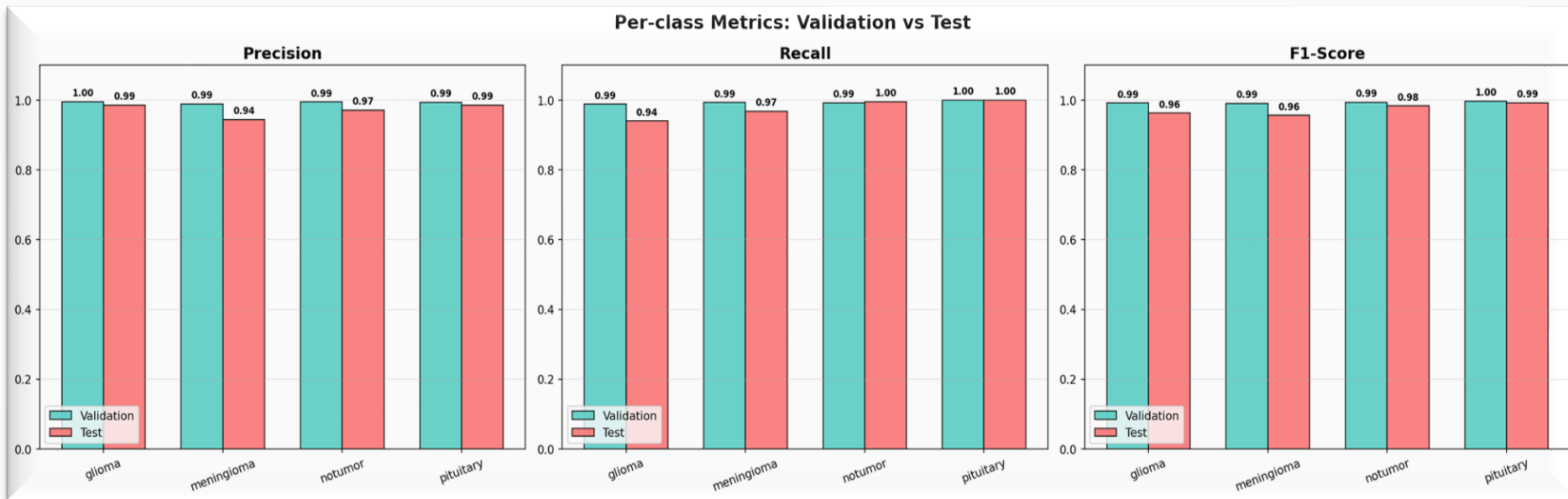
Test Vs Validation

Primary evaluation on Mendeley Brain Tumor MRI dataset



Test Vs Validation

Primary evaluation on Mendeley Brain Tumor MRI dataset



Cross-Dataset Generalization

Validated on 3 independent external datasets without re-training

**Mendeley
(Primary)**



99.22%

12,064 MRI images — (T1 , 4 classes)

**Kaggle
Brain MRI**



98.38%

Independent public dataset (T1 , 4 classes)

Br35H - Fine Tuning



90.60% - 98.83%

Binary classification subset (T2 , 2 classes)

Figshare



89.36% - 99.35%

Different MRI protocol — (T1 , 3 classes)

Strong generalization across diverse datasets confirms robustness — not dataset overfitting.

Ablation Study

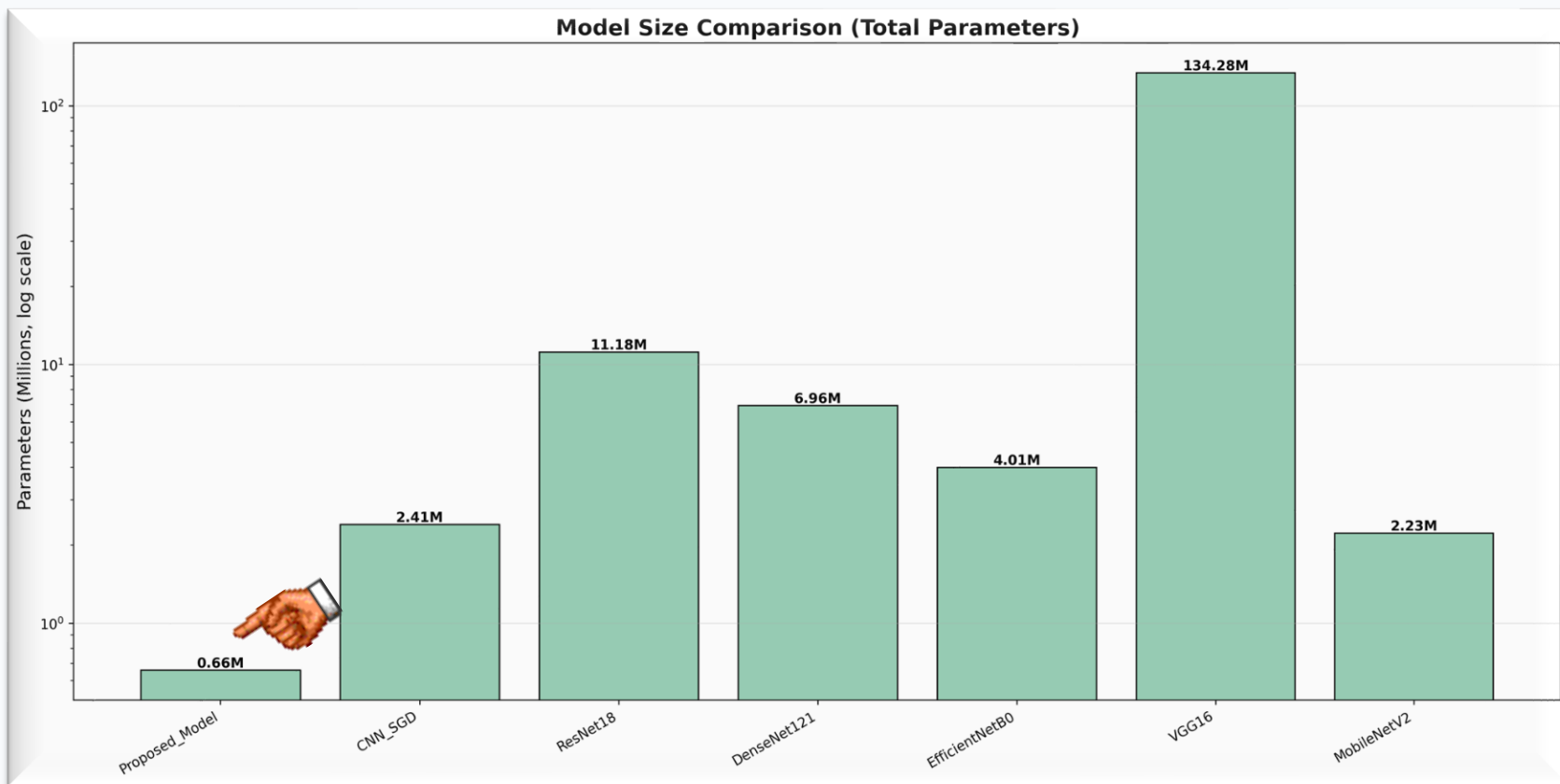
Validating the contribution of each proposed module – 50 Epochs

Each module was individually removed and re-evaluated to confirm necessity.

Configuration	Accuracy	Δ vs Baseline	Params	Best
Baseline CNN (no modules)	94.30%	–	405.6K	–
+ MAFE only	95.10%	+0.80%	410.6K	–
+ MMSP only	95.95%	+1.65%	651.5K	–
+ CAIP only	95.75%	+1.45%	408.4K	–
MAFE + CAIP (no MMSP)	96.53%	+0.88%	–	–
MAFE + MMSP (no CAIP)	98.08%	+2.43%	–	–
MMSP + CAIP (no MAFE)	98.29%	+2.64%	–	–
Full Model (MAFE+MMSP+CAIP) – 50 epochs 	98.34%	+2.69%	659,228	✓
Full Model (MAFE+MMSP+CAIP) – 300 epochs (FINAL)	99.22%	+4.92%	659,228	✓

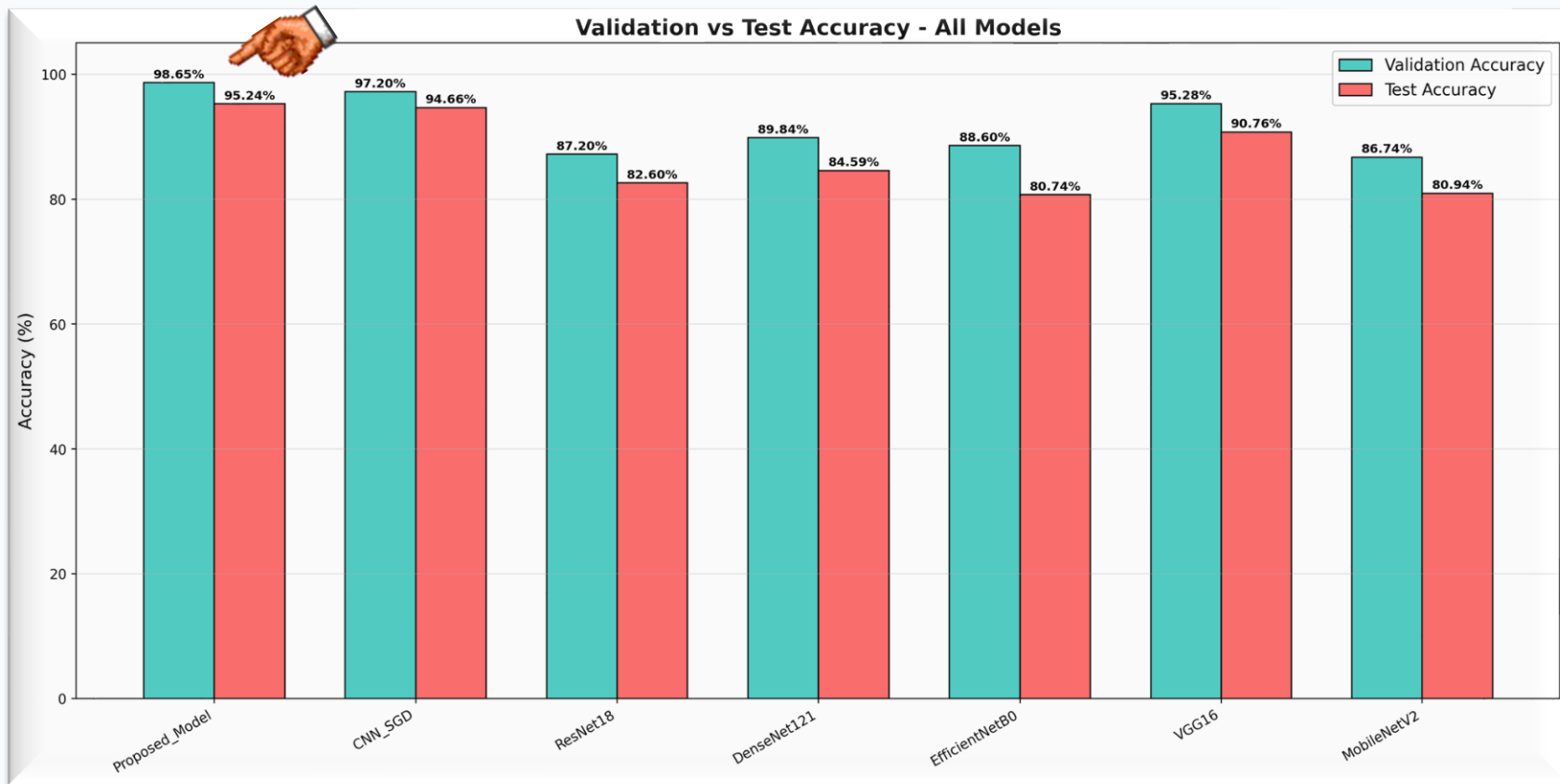
Comparative Analysis (50 Epochs)

Modified CNN vs. 6 competing algorithms



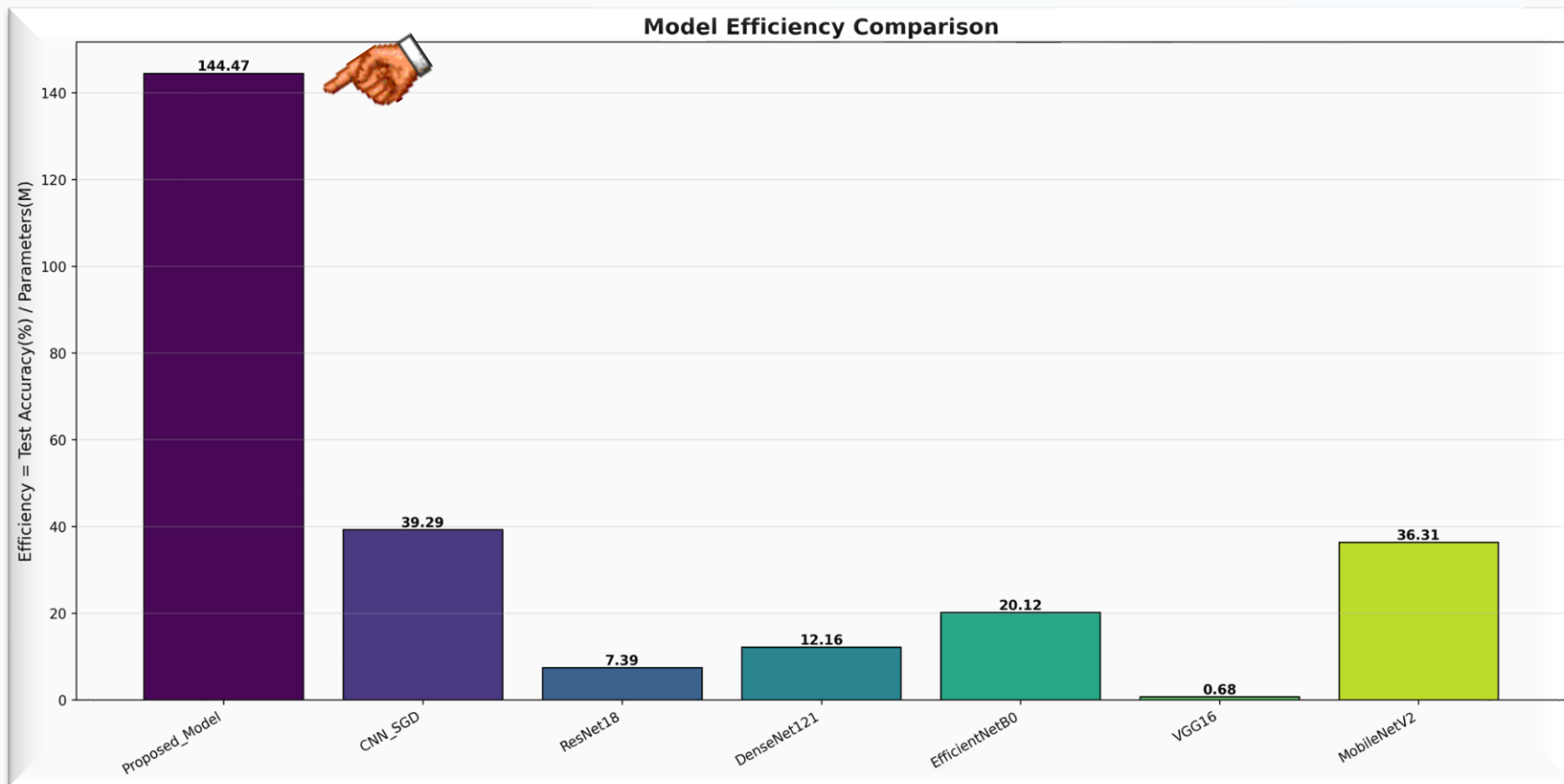
Comparative Analysis (50 Epochs)

Modified CNN vs. 6 competing algorithms



Comparative Analysis (50 Epochs)

Modified CNN vs. 6 competing algorithms

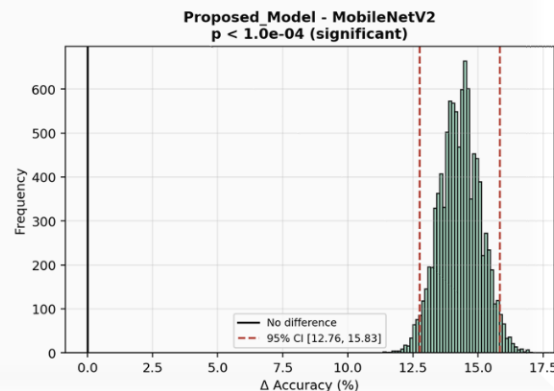
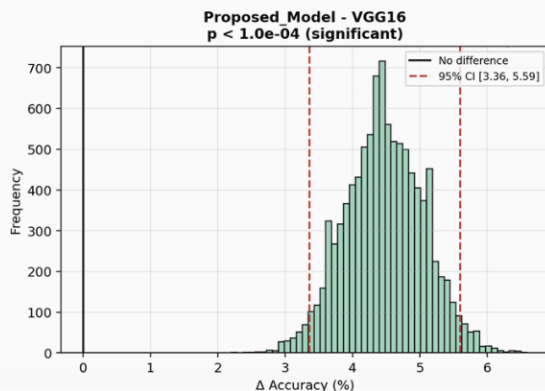
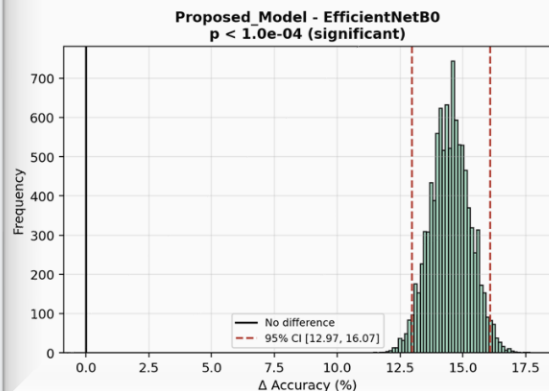
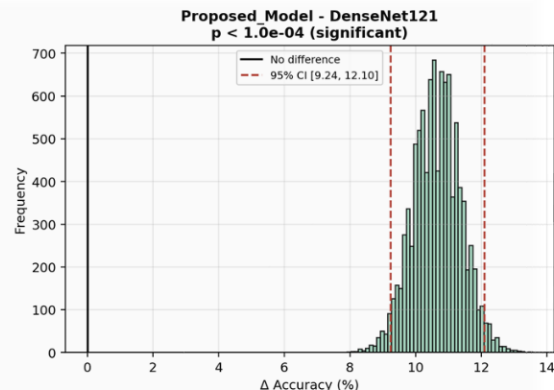
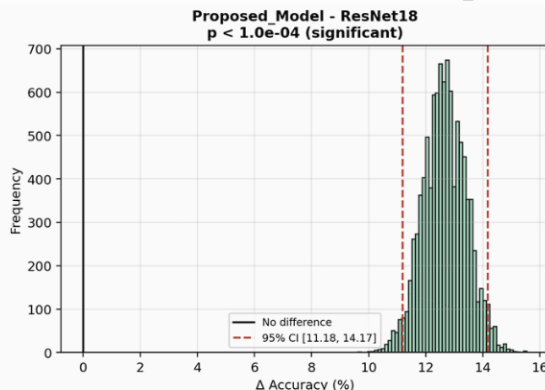
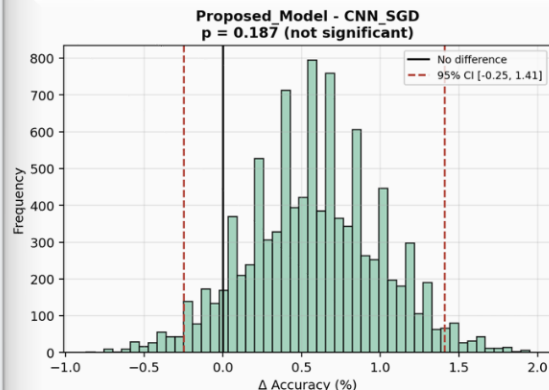


Comparative Analysis (50 Epochs)

Modified CNN vs. 6 competing algorithms (Bootstrap)



Bootstrap Paired Difference in Accuracy: Proposed_Model vs Baselines

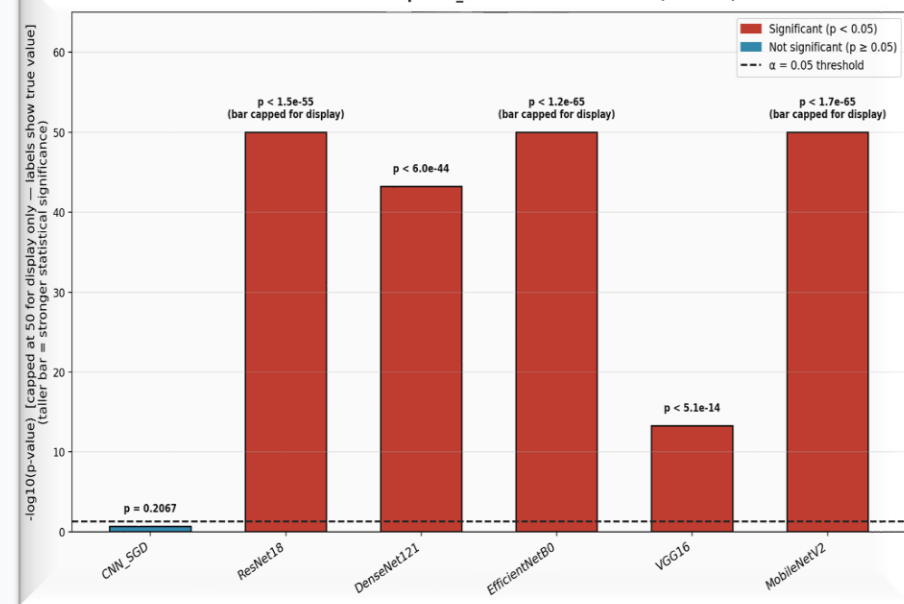


Comparative Analysis (50 Epochs)

Modified CNN vs. 6 competing algorithms (McNemar)

Algorithm

McNemar's Test: Proposed_Model vs Each Baseline (Test Set)



McNemar's Test — Pairwise p-values (Test Set)
(darker/warmer = more statistically significant, $p < 0.05$)



Three independent training runs

Primary evaluation on Mendeley Brain Tumor MRI dataset

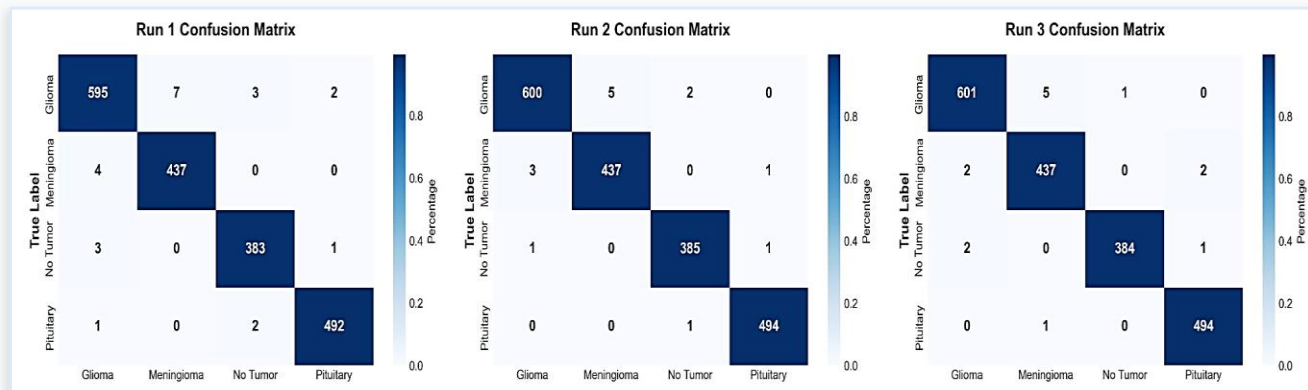
Run 1	
Accuracy	98.81%
Precision	98.81%
Recall	98.80%

Run 2	
Accuracy	99.27%
Precision	99.27%
Recall	99.27%

Run 3	
Accuracy	99.27%
Precision	99.27%
Recall	99.27%

Standard deviation across runs: $\pm 0.22\%$ (accuracy), with precision, recall, and F1-score all showing a standard deviation of $\pm 0.22\%$ as well — confirms the model is stable and reproducible across different random initializations

Confusion Matrix (Run 1 , Run 2 , Run 3)



Comparative Analysis with State-of-the-Art

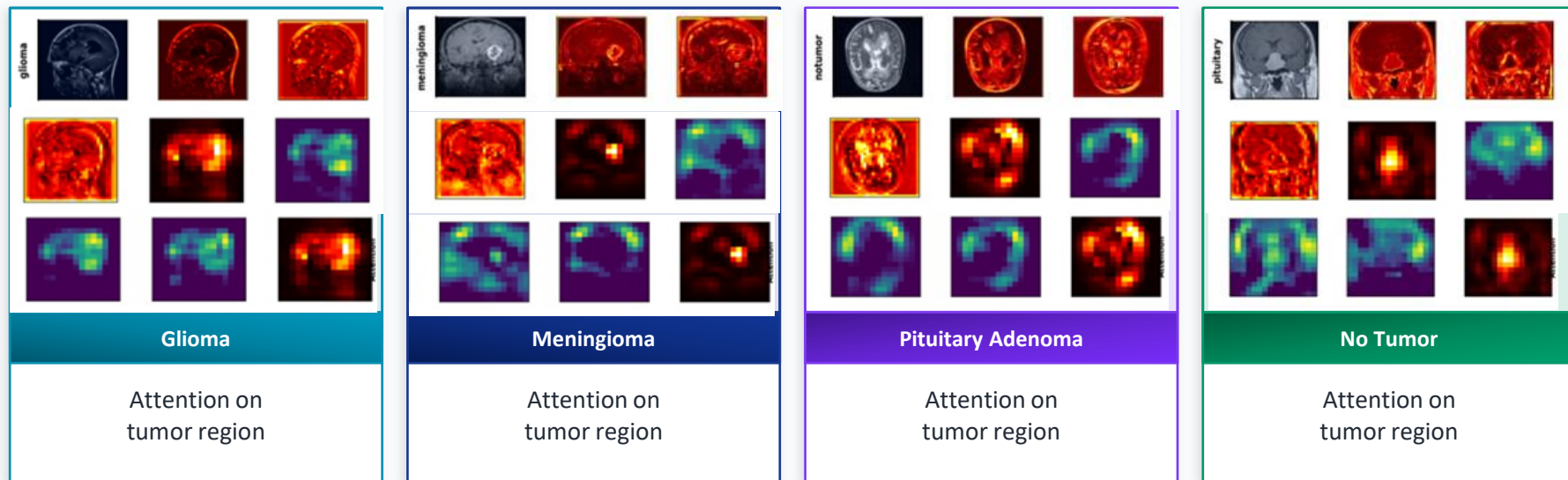
References	Year	Method	Datasets	Accuracy	Params	Efficiency
Proposed Model	2026	Modified CNN (MAFE+MMSP+CAIP)	Mendeley	99.22%	659K	150.51
Khan et al. (2025)	2025	Hybrid CE-EEN-B0-ResGANet	Multi-source	99.11%	15,000K	6.61
Zahoor et al. (2024)	2024	Res-BRNet	Multi-source	98.22%	8,200K	11.98
Khan & Auvée (2024)	2024	Resource-efficient CNN	Br35H	98.09%	1,100K	89.17
Ilani et al. (2025)	2025	U-Net CNN hybrid	T1-weighted	98.56%	31,031K	3.18
Liu & Zhao (2024)	2024	DenseNet201	BRATS	98.12%	20,000K	4.91

* Efficiency metric: Accuracy% / Parameters(M), Higher is better.

Visual Explainability — Grad-CAM

Model attention aligns with clinically relevant tumor regions

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to all four tumor classes to verify that the model attends to the correct MRI regions — matching expert radiologist focus areas.

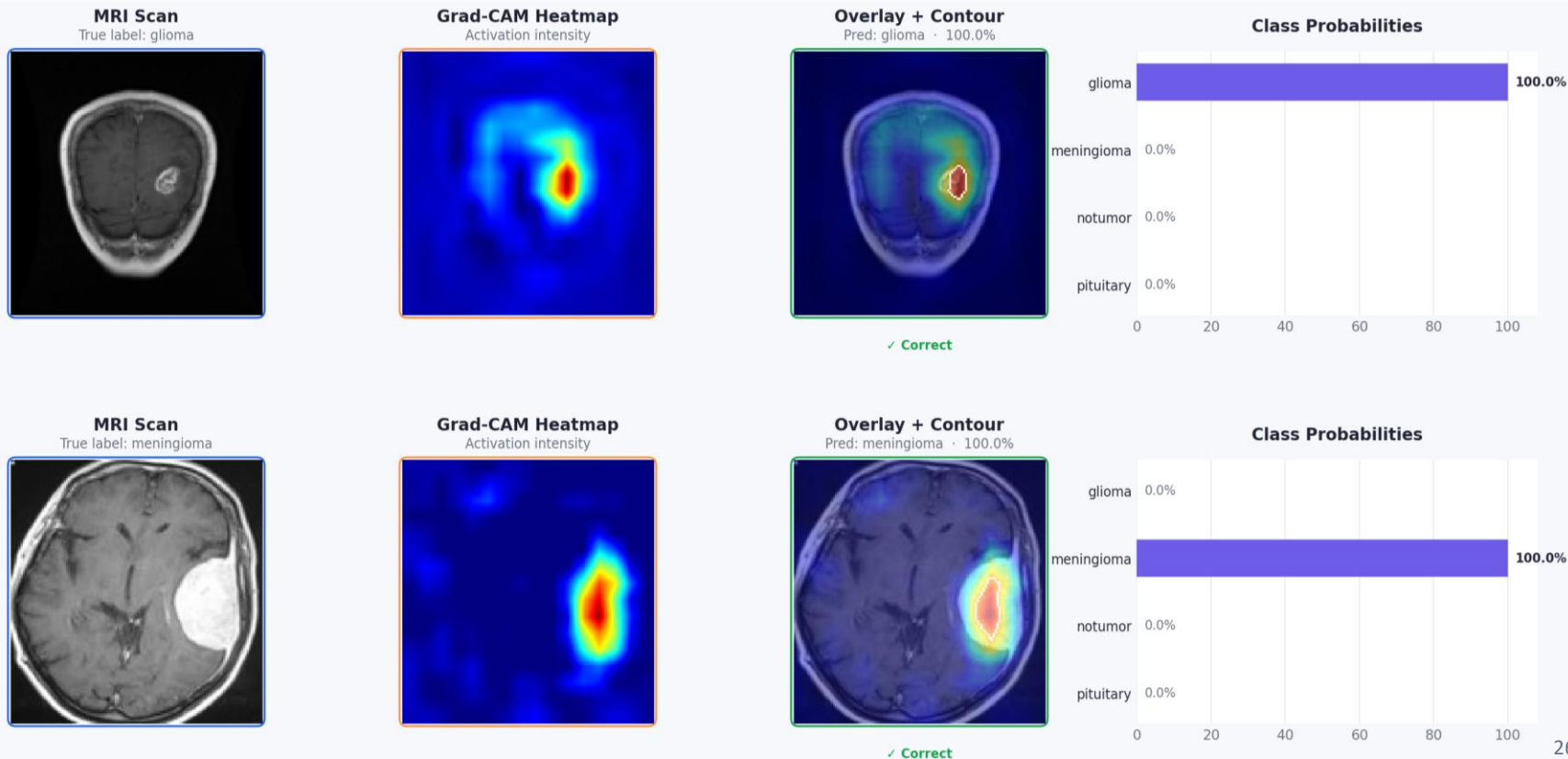


Clinical Significance:

Grad-CAM confirms that the model's decisions are clinically interpretable — a key requirement for regulatory acceptance and physician trust in AI-assisted diagnosis. Activation maps highlight tumor boundaries and lesion cores consistent with radiological standards.

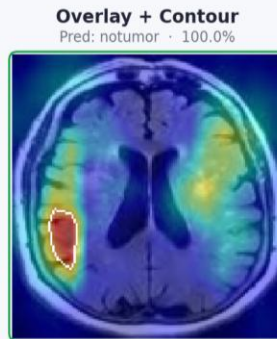
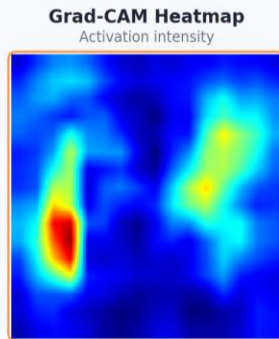
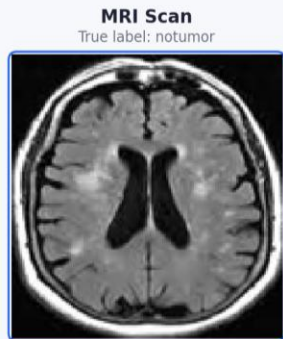
Visual Explainability — Grad-CAM

Grad-CAM Visualization Across the 4 Brain Tumor Classes

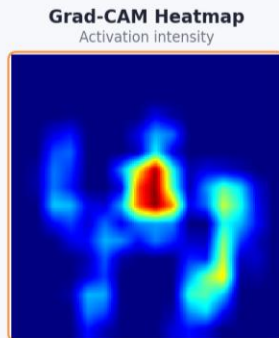
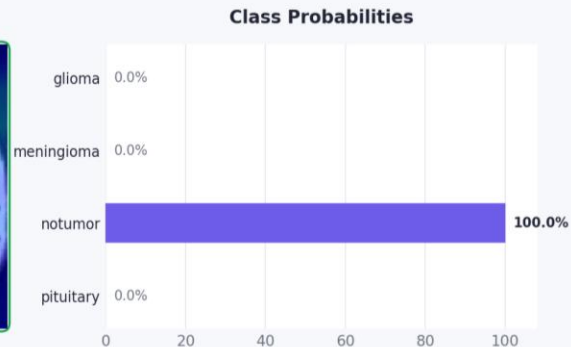


Visual Explainability — Grad-CAM

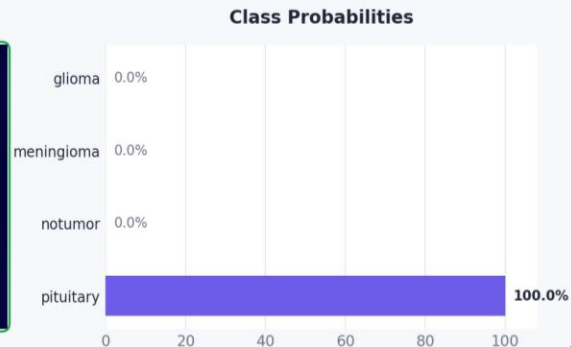
Grad-CAM Visualization Across the 4 Brain Tumor Classes



✓ Correct



✓ Correct



Conclusions & Key Contributions

Summary of achievements and contributions

①

Ultra-Lightweight Modified CNN

99.22% accuracy with only 659,228 parameters — ~98% fewer than U-Net (47-fold reduction). Parameter efficiency score of 150.51.

②

MAFE Module

Sequential spatial-then-channel attention, reversing CBAM's channel-then-spatial ordering, enriched with joint spatial attention from SE networks.

③

MMSP Module

Three-pathway multi-resolution fusion — the primary driver of performance gains (largest individual contribution among the three modules).

④

CAIP Module

Sigmoid-gated content-adaptive pooling that learns local optimal blend coefficients dynamically (+1.45% accuracy, additive ablation).

⑤

Empirical Validation

Demonstrated that architectural co-design for task-specific demands outperforms raw parameter accumulation.

Future Work & Recommendations

- Multimodal fusion — combine T1, T2, and FLAIR MRI sequences in a single model
- Federated learning — privacy-preserving training across hospital networks
- Tumor segmentation & grading — extend beyond classification to pixel-level localization
- Prospective clinical validation — partner with hospitals for real-world trials
- Regulatory approval pathway — align with FDA/CE medical device standards
- Further compression — knowledge distillation for even smaller edge deployment

No. 6

Date: 01/03/2026

To: Mahmoud S. Atheeb¹ and Baraa I. Farhan²

Address: ^{1,2}Department of Computer, College of Education for Pure Sciences, Wasit University, IRAQ

Acceptance Letter

Dear Mahmoud S. Atheeb

Re: "Optimized Lightweight Convolutional Neural Network Architecture for Multi-Class Brain Tumor Classification from Magnetic Resonance Imaging"

I am very pleased to inform you that the above-mentioned paper has been accepted for publication in our journal, Wasit journal for pure sciences and will be published in Volume 5, Issue 3. This volume will begin publication in 30 September 2026.

Yours sincerely,



Prof. Dr. Ali Al-Hachami
Editor in Chief
WJPS
Print-ISSN: 2790-5233
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Lightweight Deep Learning Models for Brain Tumor Classification

Mahmoud Shmrar Atheeb¹ and Baraa Ismael Farhan²

¹Ministry of Education, Department of Computer Science, College of Education for Pure Sciences, Wasit University, 52001 Al-Kut, Iraq

²Ministry of Higher Education, Department of Computer Science, College of Education for Pure Sciences, Wasit University, 52001 Al-Kut, Iraq

Mahmoud shmrar88@gmail.com, bfarhan@uowasit.edu.iq

Keywords: Brain Tumor Classification, Convolutional Neural Networks, Attention Mechanisms, Medical Image Analysis.

Abstract: Differentiating brain tumours by MRI using computer algorithms remains a huge challenge in clinical neuro-oncology and medical imaging area. In our work, we proposed a new ultra-lightweight convolutional neural network (CNN) deep structure for practical use in clinical setting. Our CNN embeddings three private self-intelligent modules: (1) a Micro Adaptive Feature Extractor applied spatial-channel attention to whole image to have a dynamic feature refinement; (2) a Micro Multi-Scale Processor that replicates expert radiologist diagnostic manoeuvres by enabling learned multi-resolution feature combination; and (3) a Context-Aware Intelligent Pooling that offers content-based flexible pooling strategies. The complete Mendeley Brain Tumour MRI dataset (n=12,064 images) was used to test each approach, and resulted with an excellent classification accuracy of 99.22%. Such Mendeley Brain Tumour MRI dataset had an extremely tiny footprints of only 659.228 trainable parameters, which was 80% less intricate than the other architectures such as U-Net. The model demonstrates that it could be effective on three independent external validation datasets: Kaggle (99.24%), Epixare (87.47%) and Bc35H (98.33%). This demonstrates that the model can be applied across datasets. We achieved several essential features needed for clinical deployment for reducing inference latency (12ms), a smaller model footprint (2.6 MB), Achieved good results given the system complexity (network size, operation count, parameter size) and the frugal memory utilization (4.4 MB RAM). Ablation analysis proved that the intelligent pooling module contributed to accuracy increase up to 2.18%. This study represents a significant step forward in parameters-efficient deep learning for medical imaging, enabling feasible deployment on mobile computing platforms, edge computing devices, and in norm constrained healthcare centres around the world.

1 INTRODUCTION

Brain tumors are one of the hardest things to pick up in modern day neuro-oncology. Healthcare structures worldwide are becoming increasingly, and increasingly burdened by the management of brain tumors. They are increasing in incidence and thus has implications for morbidity and mortality of the population whatever the birth place or race. The most sensitive modality for the detection, diagnosis and differential diagnosis of brain tumors is now with MRI. Primarily because of the superior soft-tissue contrast resolution it provides as well as the varied imaging options it offers [1]. However, imaging techniques such as MRI are not that useful in determining tumor histology. This is because

observers may have subjective interpretations of the results and a significant few experienced neuro-radiologists would have different biases when interpreting the effects [2]. It is much more difficult to determine the various types of intracranial neoplasms individuals suffer from due to the wide range of types there are. There are four main categories of brain tumors which are: a) Gliomas, these are the most common types of primary brain cancers and where the grades are precise determining factor in sorts of treatment regimen to adhere to [3]; b) Meningiomas, which are derived from the dural - arachnoid matter and represent tumors of a less aggressive nature; c) Pituitary gland adenomas, which are derived from the antero-pituitary and are common lesions seen in endocrine pathologies and d) normal brain tissue - the most valuable control group both in

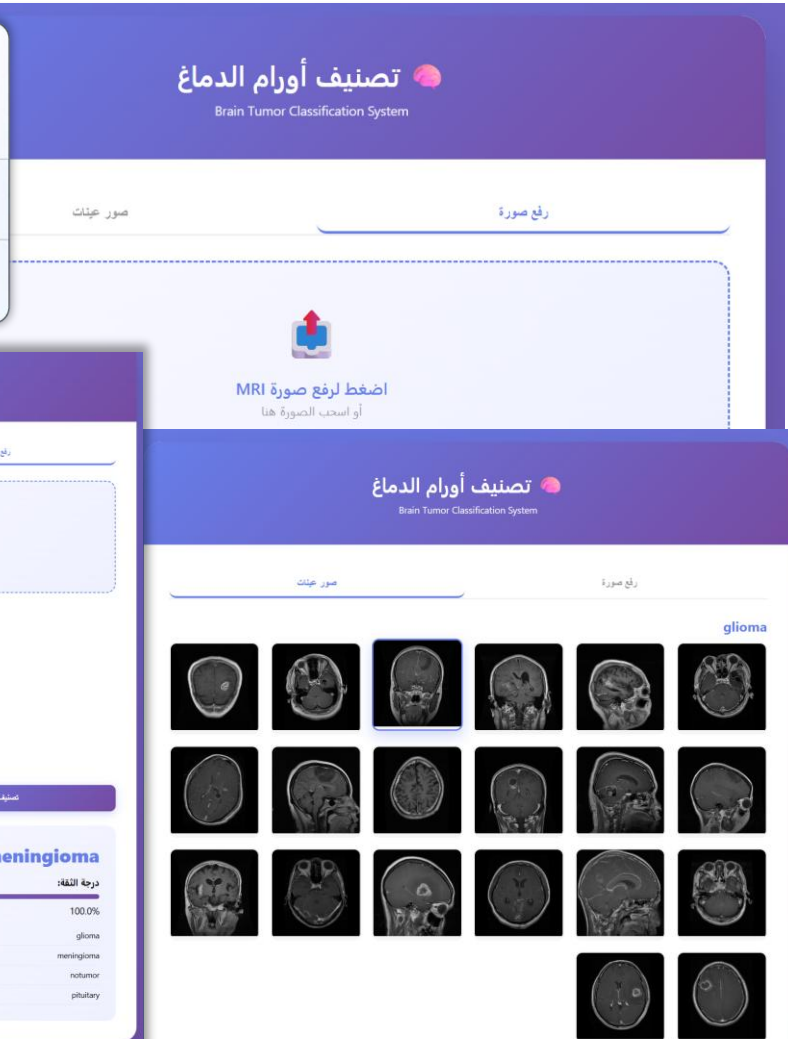
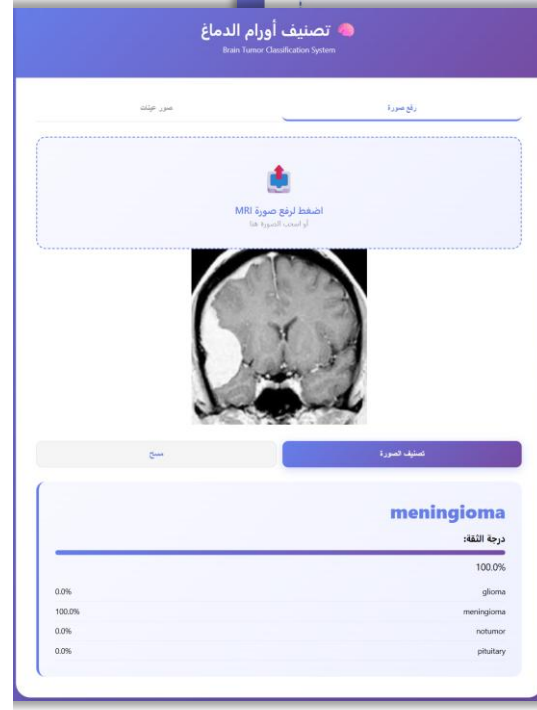
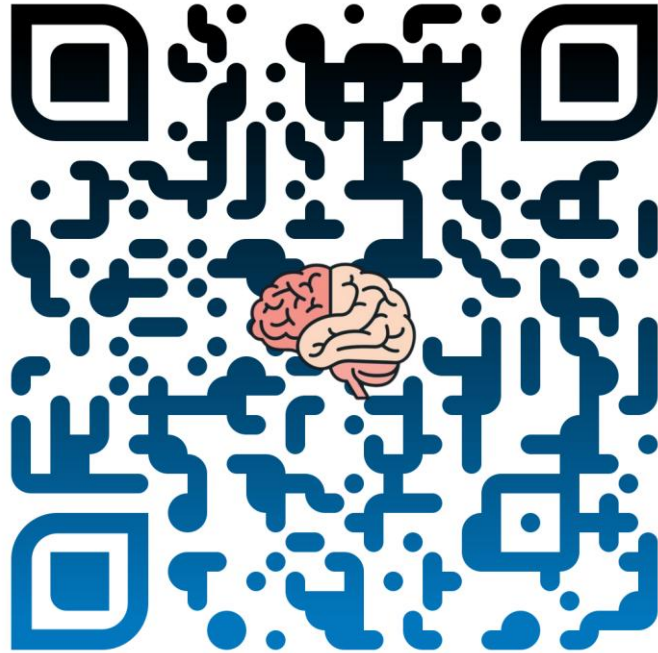


Thank You

Questions & Discussion Welcome

لتجربة النموذج الآن امسح الباركود

http://46.202.188.132/brain_tumor/



بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

تجري بعونه تعالى مناقشة رسالة طالب الماجستير

محمود شمران عذيب

الموسومة بـ:

An Efficient Deep Learning Approach with Modified CNN for Multi-Class Brain Tumor Classification



القاعة المركزية للتعليم المستمر
جامعة واسط



الساعة التاسعة صباحاً
(09:00 AM)



يوم الأحد الموافق
2026/08/16



وتتألف لجنة المناقشة من الأساتذة المدرجة أسماؤهم أدناه :

ت	اللقب العلمي	الاسم التدريسي	الشهادة	التخصص الدقيق	مكان العمل	المنصب
1	أستاذ	أ.د. بلال اسماعيل خليل	دكتوراه	ذكاء اصطناعي	جامعة الأنبار / كلية علوم الحاسوب وتكنولوجيا المعلومات	رئيساً
2	أستاذ مساعد	أ.م.د. جعفر صادق قطيف	دكتوراه	شبكات الحاسوب	جامعة واسط / كلية التربية للعلوم الصرفة	عضواً
3	مدرس	م.د. علي فاضل راشد	دكتوراه	ذكاء اصطناعي	جامعة واسط / كلية التربية للعلوم الصرفة	عضواً
4	أستاذ مساعد	أ.م.د. براء اسماعيل فرحان	دكتوراه	علوم وهندسة الحاسبات	جامعة واسط / كلية التربية للعلوم الصرفة	عضواً ومشرفاً



نتشرف بحضوركم ومشاركتكم في إنجاح هذه المناقشة العلمية

